Chapter 38

Diffraction Patterns and Polarization (nhiễu xạ và phân cực)

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Diffraction and Polarization

Diffraction can be described only with a wave model for light.

A diffraction pattern occurs when the light from an aperture is allowed to fall on a screen .

- The features of this diffraction pattern can be investigated.

Under certain conditions transverse waves with electric field vectors in all directions can be polarized in various ways .

- Only certain directions of the electric field vectors are present in the polarized wave.

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Diffraction

Light of wavelength comparable to or larger than the width of a slit spreads out in all forward directions upon passing through the slit.

This phenomena is called *diffraction.*

- This indicates that light spreads beyond the narrow path defined by the slit into regions that would be in shadow if light traveled in straight lines.

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Diffraction Pattern (Ảnh nhiễu xạ)

A single slit placed between a distant light source and a screen produces a **diffraction pattern.**

- It will have a broad, intense central band
  - Called the **central maximum (cực đại giữa, cực đại trung tâm)**
- The central band will be flanked by a series of narrower, less intense secondary bands.
  - Called **side maxima** or **secondary maxima (cực đại phụ)**
- The central band will also be flanked by a series of dark bands.
  - Called **minima (cực tiểu)**

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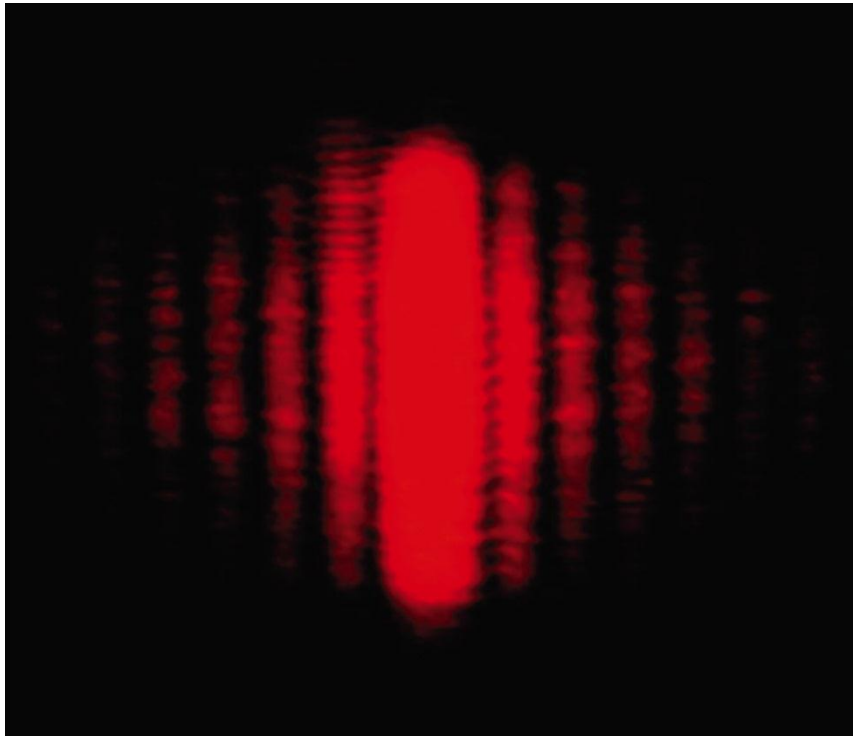
Diffraction Pattern, Single Slit

The diffraction pattern consists of the central maximum and a series of secondary maxima and minima.

The pattern is similar to an interference pattern.



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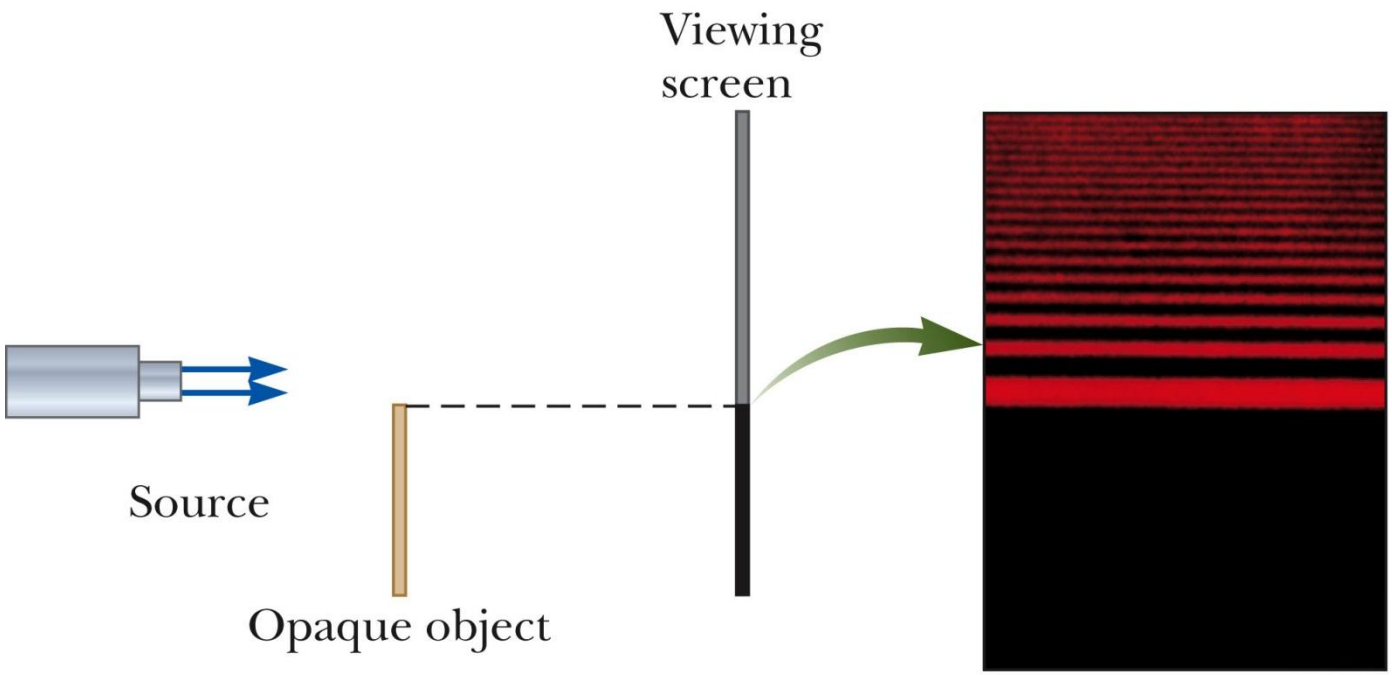
Diffraction Pattern, Object Edge

This shows the upper half of the diffraction pattern formed by light from a single source passing by the edge of an opaque object.

The diffraction pattern is vertical with the central maximum at the bottom.



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Confirming Wave Nature

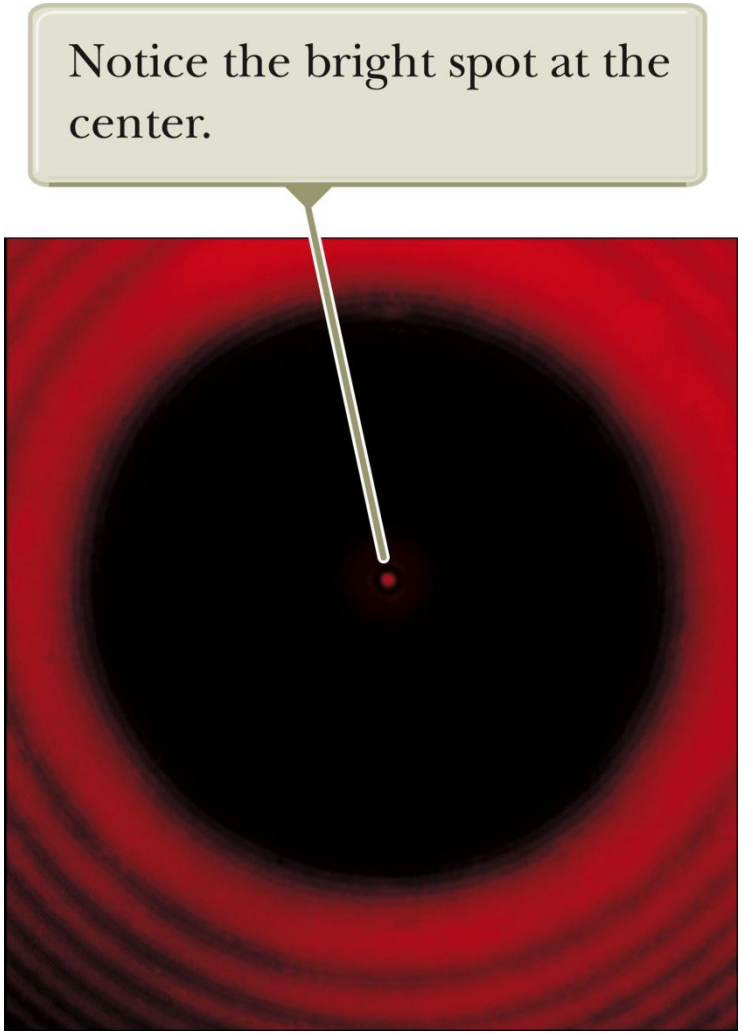
Ray optics would predict a dark spot in the center.

Wave theory predicts the presence of the center spot.

There is a bright spot at the center.

- Confirms wave theory

The circular fringes extend outward from the shadow's edge.



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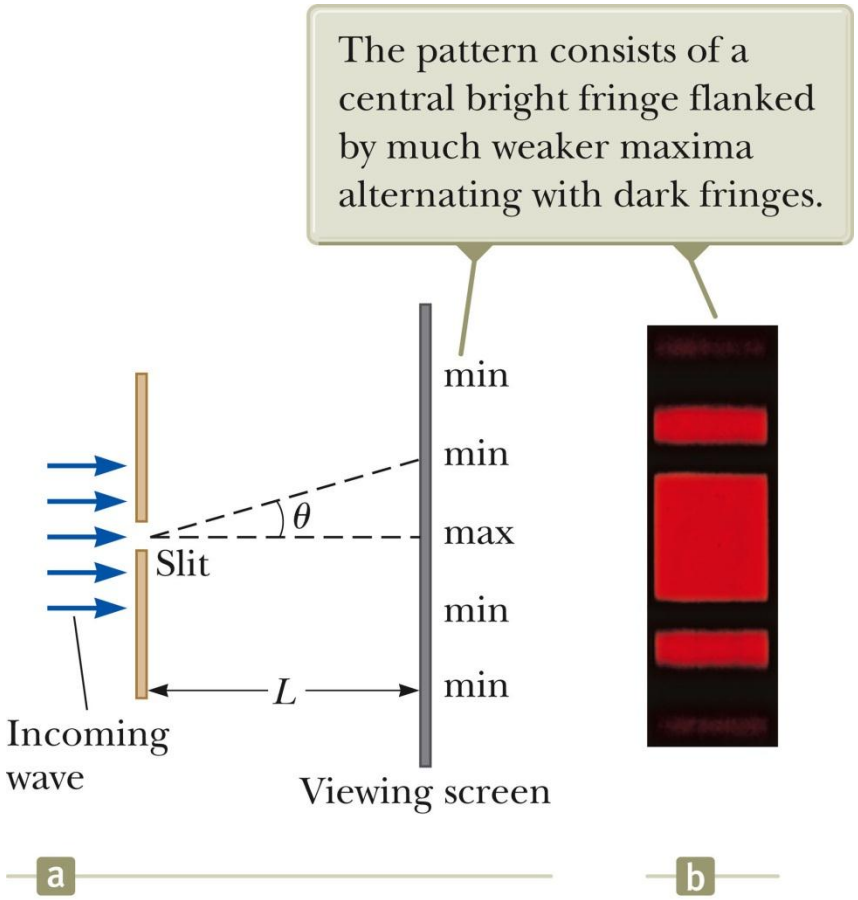
Fraunhofer Diffraction Pattern

A **Fraunhofer diffraction pattern** occurs when the rays leave the diffracting object in parallel directions.

- Screen very far from the slit
- Could be accomplished by a converging lens



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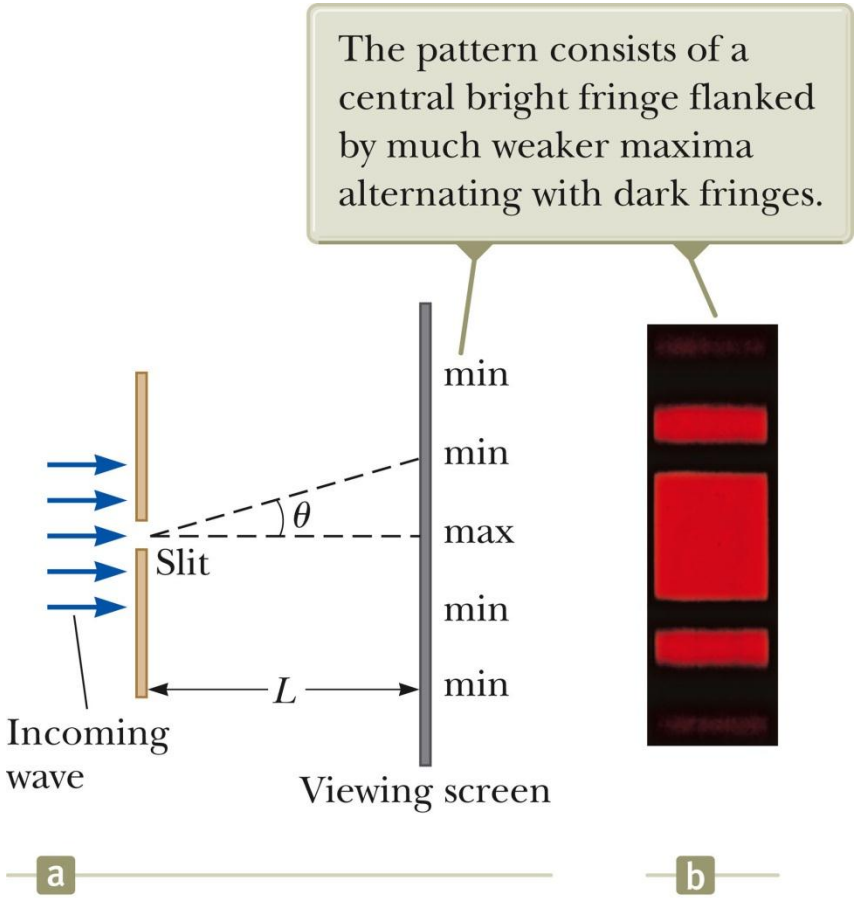
Fraunhofer Diffraction Pattern Photo

A bright fringe is seen along the axis ($\theta = 0$).

Alternating bright and dark fringes are seen on each side.



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Diffraction vs. Diffraction Pattern

Diffraction refers to the general behavior of waves spreading out as they pass through a slit.

A *diffraction pattern* is actually a misnomer that is deeply entrenched.

- The pattern seen on the screen is actually another *interference* pattern.
- The interference is between parts of the incident light illuminating different regions of the slit.

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Single-Slit Diffraction

The finite width of slits is the basis for understanding Fraunhofer diffraction.

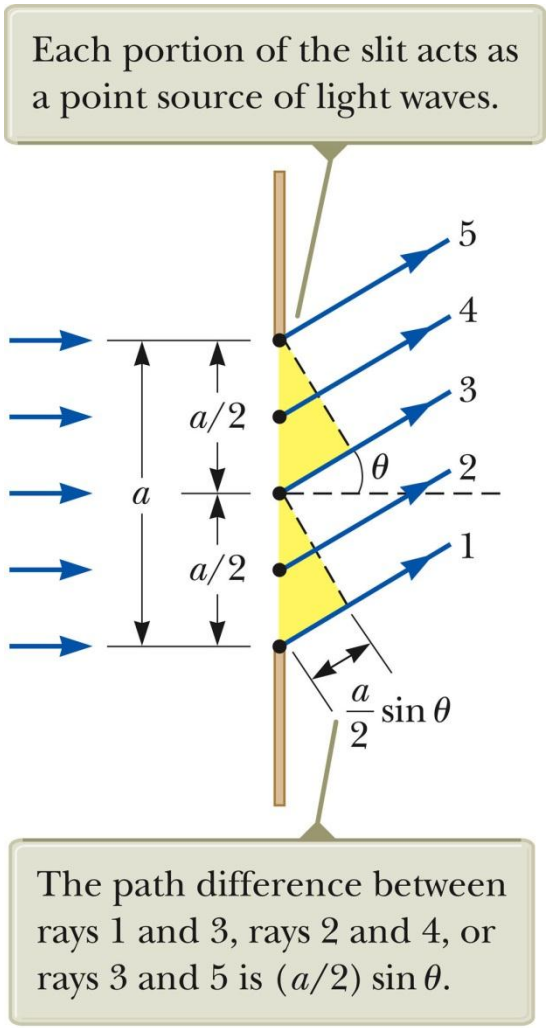
According to Huygens's principle, each portion of the slit acts as a source of light waves.

Therefore, light from one portion of the slit can interfere with light from another portion.

The resultant light intensity on a viewing screen depends on the direction θ.

The diffraction pattern is actually an interference pattern.

- The different sources of light are different portions of the single slit.



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Single-Slit Diffraction, Analysis

All the waves are in phase as they leave the slit.

Wave 1 travels farther than wave 3 by an amount equal to the path difference.

- $(a/2) \sin \theta$

If this path difference is exactly half of a wavelength, the two waves cancel each other and destructive interference results.

In general, destructive interference occurs for a single slit of width a when $\sin \theta_{dark} = m\lambda / a$.

- $m = \pm 1, \pm 2, \pm 3, \ldots$

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Single-Slit Diffraction, Intensity

The general features of the intensity distribution are shown.

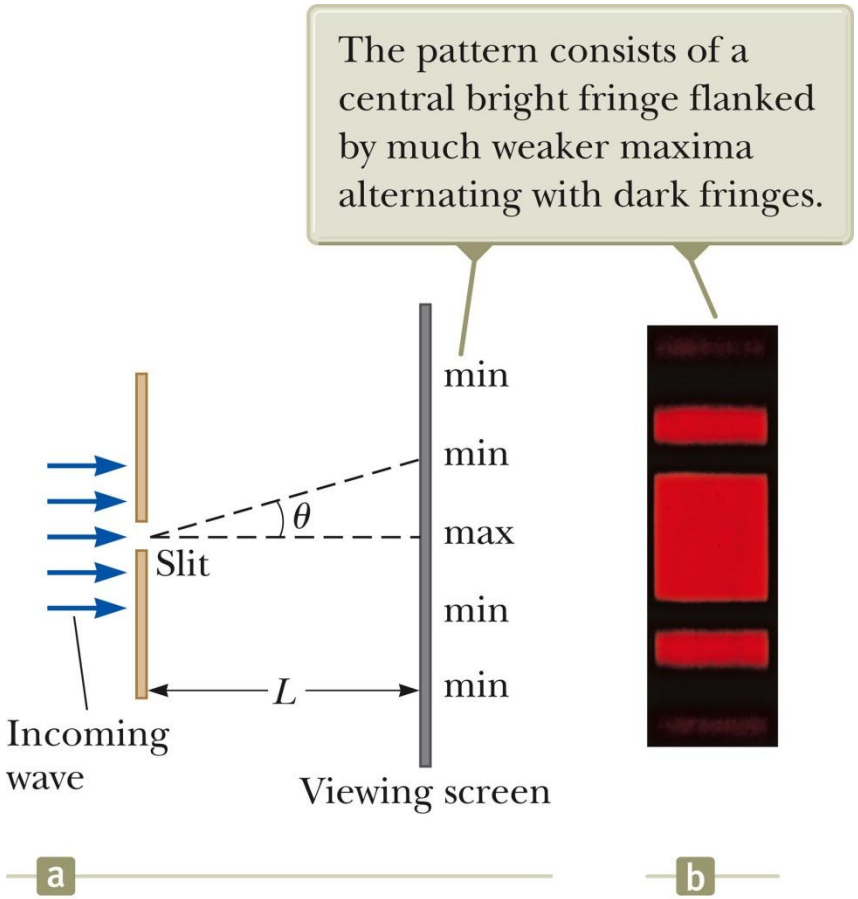
A broad central bright fringe is flanked by much weaker bright fringes alternating with dark fringes.

Each bright fringe peak lies approximately halfway between the dark fringes.

The central bright maximum is twice as wide as the secondary maxima.

There is no central dark fringe.

- Corresponds to no m = 0 in the equation



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Intensity, equation

The intensity can be expressed as

$$I = I_{max}\left[\frac{\sin(\pi a \sin\theta/\lambda)}{\pi a \sin\theta/\lambda}\right]^2$$

Minima occur at

$$\frac{\pi a \sin\theta_{dark}}{\lambda} = m\pi \quad \text{or} \quad \sin\theta_{dark} = m\frac{\lambda}{a}$$

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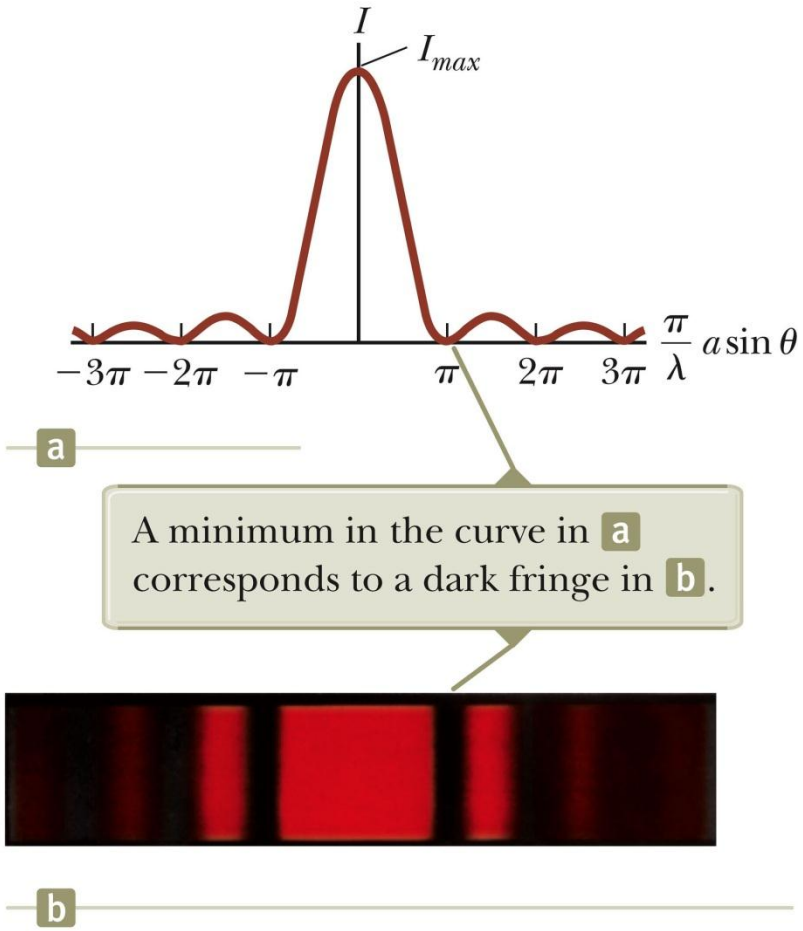
Intensity, final

Most of the light intensity is concentrated in the central maximum.

The graph shows a plot of light intensity vs. (π /λ) a sin θ.



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Intensity of Two-Slit Diffraction Patterns

When more than one slit is present, consideration must be made of

- The diffraction patterns due to individual slits
- The interference due to the wave coming from different slits

The single-slit diffraction pattern will act as an “envelope” for a two-slit interference pattern.

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Intensity of Two-Slit Diffraction Patterns, Equation

To determine the maximum intensity:

$$I = I_{\max} \cos^2\left(\frac{\pi d \sin\theta}{\lambda}\right)\left[\frac{\sin(\pi a \sin\theta / \lambda)}{\pi a \sin\theta / \lambda}\right]^2$$

- The factor in the square brackets represents the single-slit diffraction pattern.
  - This acts as the envelope.
- The two-slit interference term is the $\cos^2$ term.

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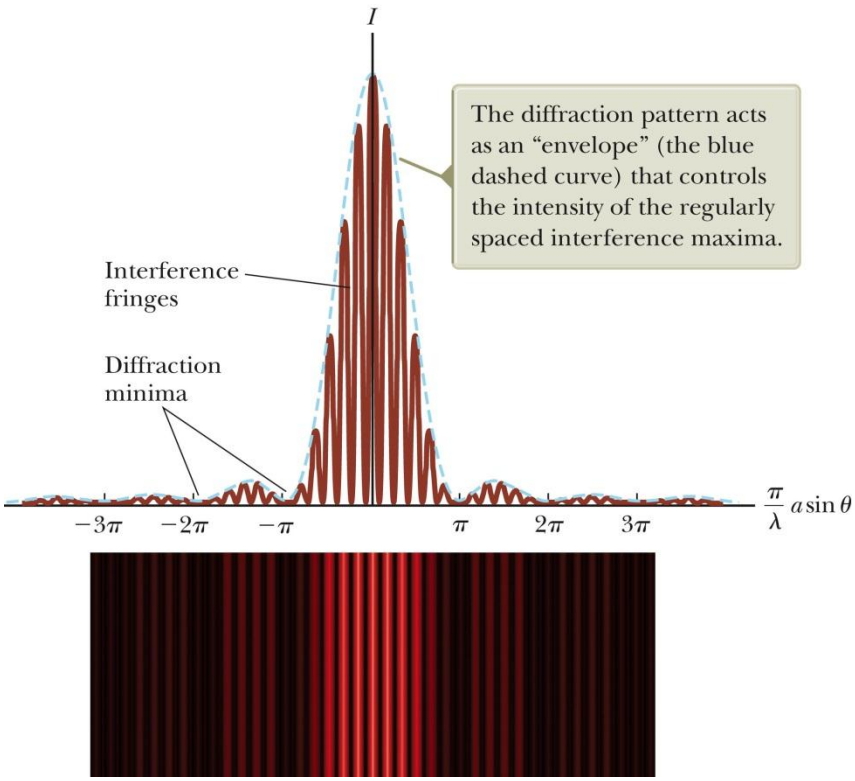
Intensity of Two-Slit Diffraction Patterns, Graph of Pattern

The broken blue line is the diffraction pattern.

The brown curve shows the $\cos^2$ term.

- This term, by itself, would result in peaks with all the same heights.
- The uneven heights result from the diffraction term (square brackets in the equation).



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Two-Slit Diffraction Patterns, Maxima and Minima

To find which interference maximum coincides with the first diffraction minimum.

$$\frac{d\sin\theta}{a\sin\theta} = \frac{m\lambda}{\lambda} \rightarrow \frac{d}{a} = m$$

- The conditions for the first interference maximum
  - $d \sin \theta = m \lambda$
- The conditions for the first diffraction minimum
  - $a \sin \theta = \lambda$

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Resolution

The ability of optical systems to distinguish between closely spaced objects is limited because of the wave nature of light.

If two sources are far enough apart to keep their central maxima from overlapping, their images can be distinguished.

- The images are said to be resolved.

If the two sources are close together, the two central maxima overlap and the images are not resolved.

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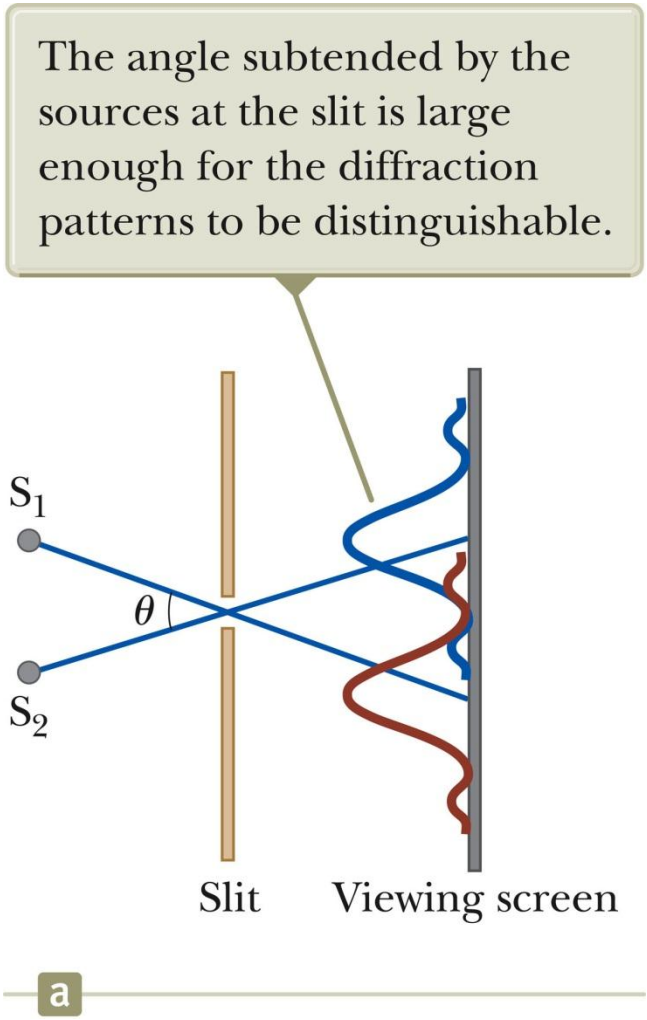
Resolved Images, Example

The images are far enough apart to keep their central maxima from overlapping.

The angle subtended by the sources at the slit is large enough for the diffraction patterns to be distinguishable.

The images are resolved.



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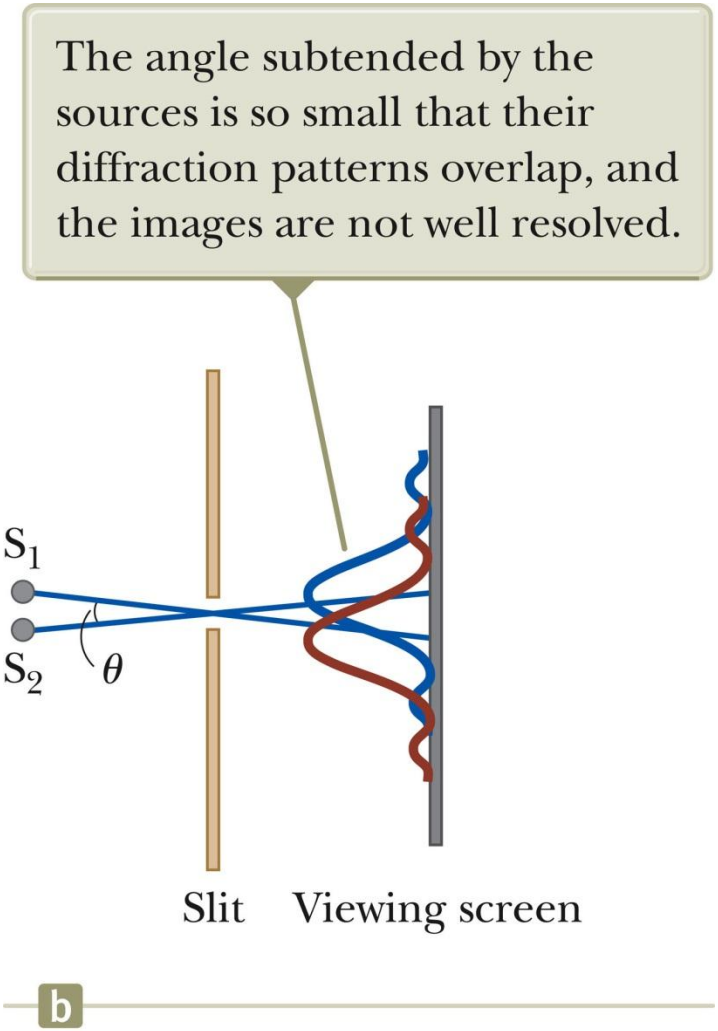
Images Not Resolved, Example

The sources are so close together that their central maxima do overlap.

The angle subtended by the sources is so small that their diffraction patterns overlap.

The images are not resolved.



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Resolution, Rayleigh's Criterion

When the central maximum of one image falls on the first minimum of another image, the images are said to be just resolved.

This limiting condition of resolution is called **Rayleigh's criterion.**

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Resolution, Rayleigh's Criterion, Equation

The angle of separation, θ_{min}, is the angle subtended by the sources for which the images are just resolved.

Since $\lambda << a$ in most situations, $\sin\theta$ is very small and $\sin\theta \approx \theta$.

Therefore, the limiting angle (in rad) of resolution for a slit of width a is

$$\theta_{min} = \lambda / a$$

To be resolved, the angle subtended by the two sources must be greater than θ_{min}.

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Circular Apertures

The diffraction pattern of a circular aperture consists of a central bright disk surrounded by progressively fainter bright and dark rings.

The limiting angle of resolution of the circular aperture is

$$\theta_{\text{min}} = 1.22\,\frac{\lambda}{D}$$

- D is the diameter of the aperture.

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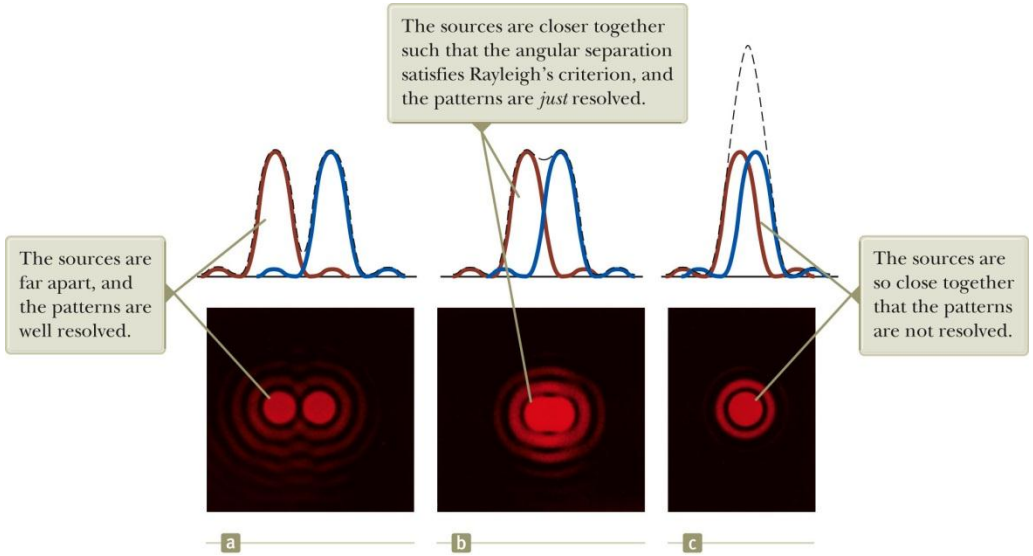
Circular Apertures, Well Resolved

The sources are far apart.

The images are well resolved.

The solid curves are the individual diffraction patterns.

The dashed lines are the resultant pattern.



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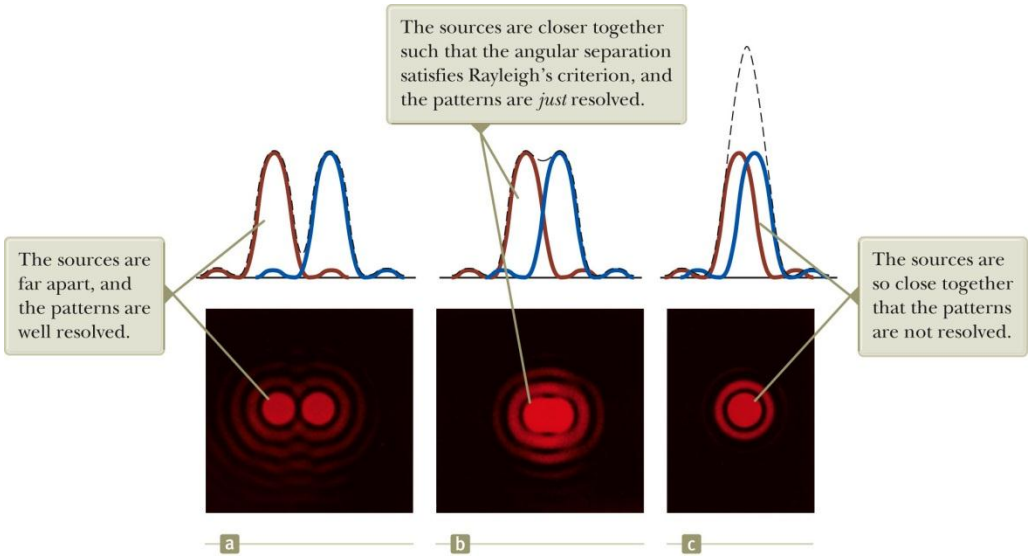
Circular Apertures, Just Resolved

The sources are separated by an angle that satisfies Rayleigh's criterion.

The images are just resolved.

The solid curves are the individual diffraction patterns.

The dashed lines are the resultant pattern.



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Circular Apertures, Not Resolved

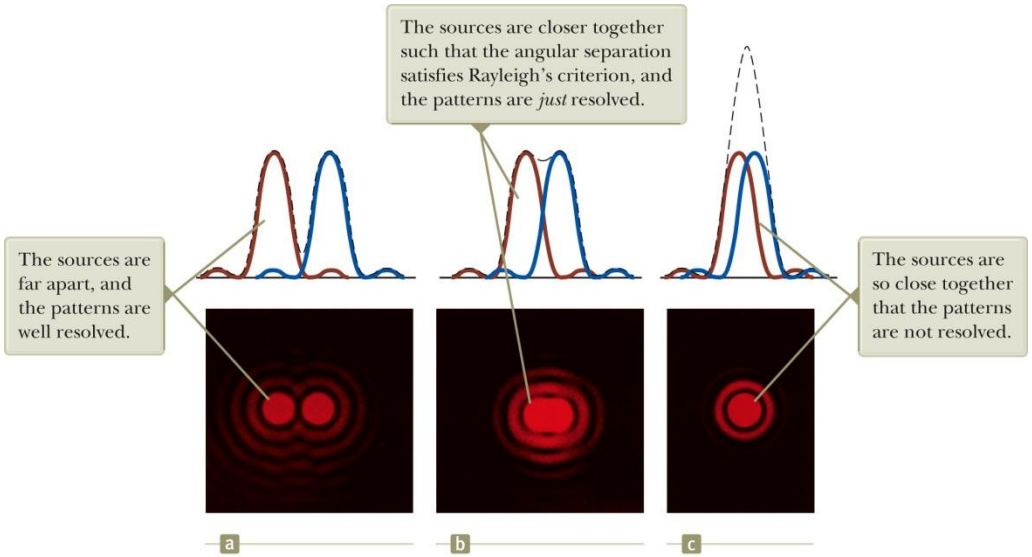
The sources are close together.

The images are unresolved.

The solid curves are the individual diffraction patterns.

The dashed lines are the resultant pattern.

The pattern looks like that of a single source.



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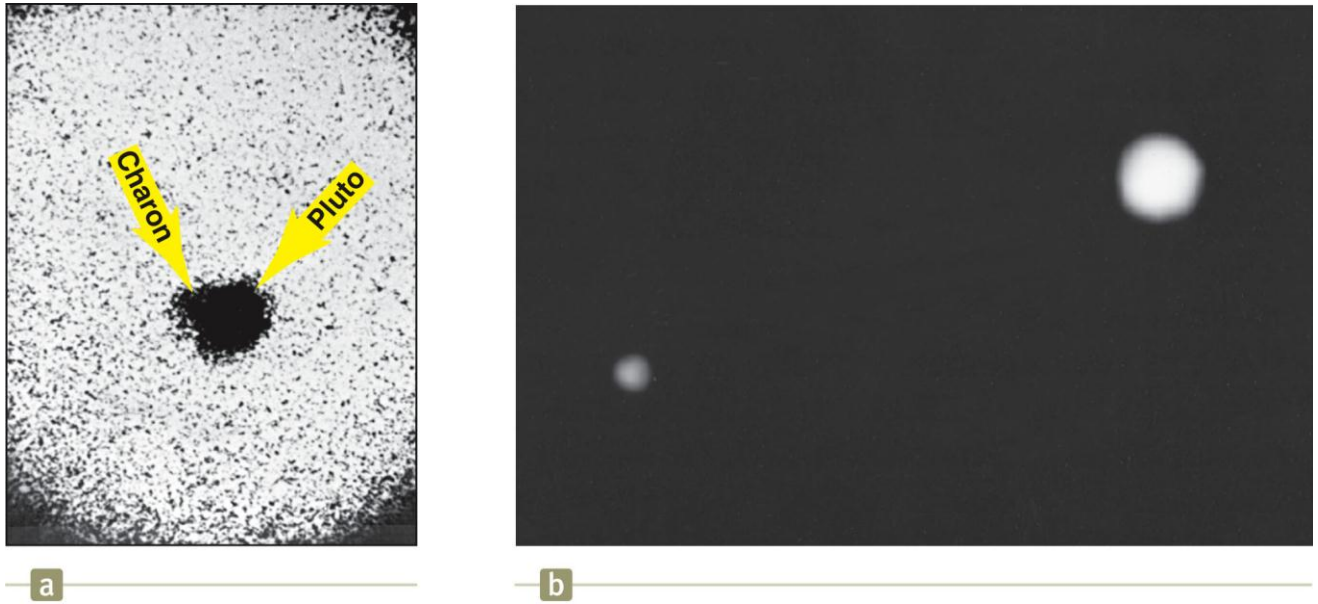
Resolution, Example



Pluto and its moon, Charon

Left: Earth-based telescope is blurred

Right: Hubble Space Telescope clearly resolves the two objects

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Diffraction Grating

The diffracting grating consists of a large number of equally spaced parallel slits.

- A typical grating contains several thousand lines per centimeter.

The intensity of the pattern on the screen is the result of the combined effects of interference and diffraction.

- Each slit produces diffraction, and the diffracted beams interfere with one another to form the final pattern.

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Diffraction Grating, Types

A *transmission* grating can be made by cutting parallel grooves on a glass plate.

- The spaces between the grooves are transparent to the light and so act as separate slits.

A *reflection* grating can be made by cutting parallel grooves on the surface of a reflective material.

- The spaces between the grooves act as parallel sources of reflected light, like the slits in a transmission grating.

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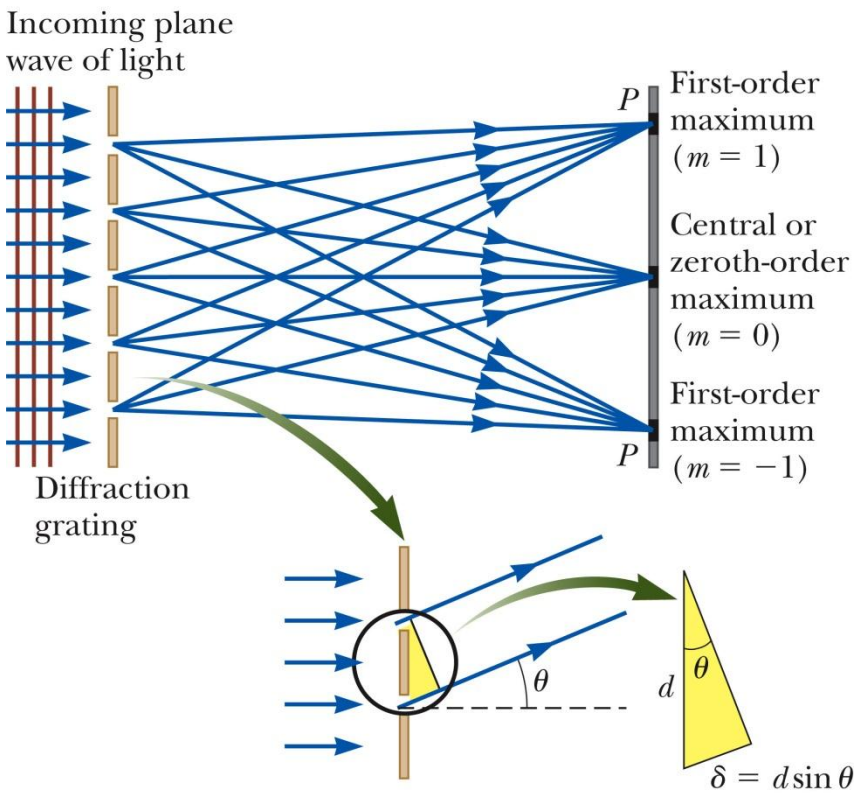
Diffraction Grating, cont.

The condition for *maxima* is

- $d \sin \theta_{bright} = m\lambda$
  - m = 0, ±1, ±2, …

The integer *m* is the *order number* of the diffraction pattern.

If the incident radiation contains several wavelengths, each wavelength deviates through a specific angle.



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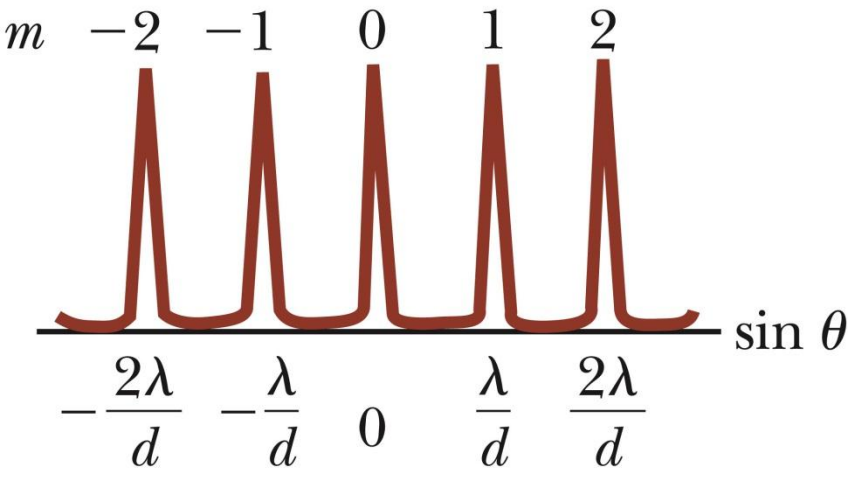
Diffraction Grating, Intensity

All the wavelengths are seen at $m = 0$.

- This is called the zeroth-order maximum.

The first-order maximum corresponds to $m = 1$.

Note the sharpness of the principle maxima and the broad range of the dark areas.



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Diffraction Grating, Intensity, cont.

Characteristics of the intensity pattern:

- The sharp peaks are in contrast to the broad, bright fringes characteristic of the two-slit interference pattern.
- Because the principle maxima are so sharp, they are much brighter than two-slit interference patterns.

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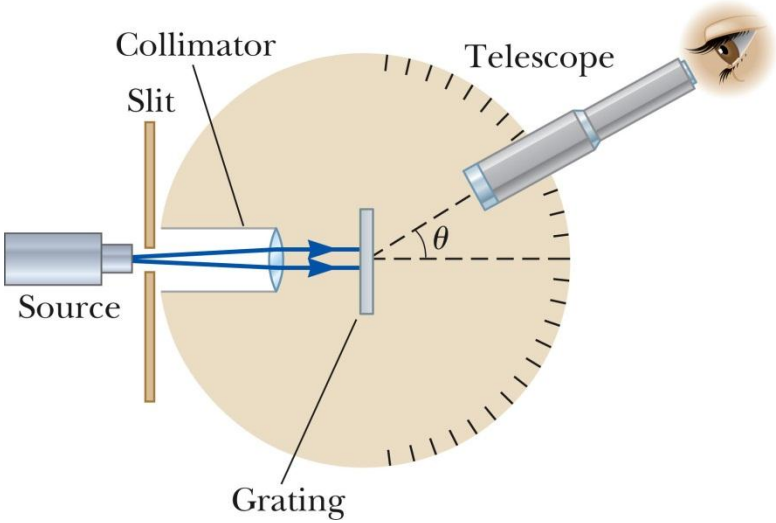
Diffraction Grating Spectrometer

The collimated beam is incident on the grating.

The diffracted light leaves the gratings and the telescope is used to view the image.

The wavelength can be determined by measuring the precise angles at which the images of the slit appear for the various orders.



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Grating Light Valve (GLV)

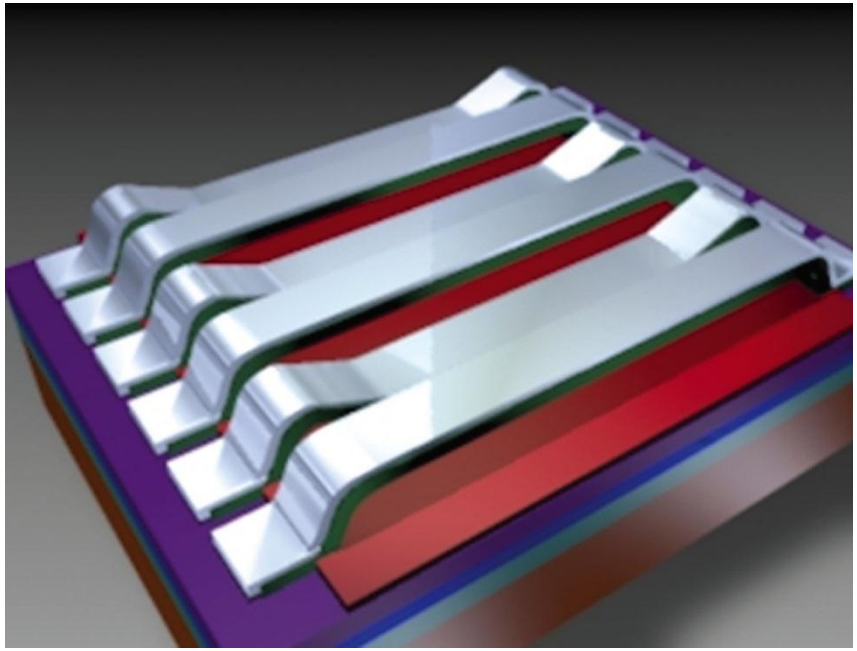
The GLV is used in some video display applications.

A GLV is a silicon microchip fitted with an array of parallel silicon nitride ribbons coated with a thin layer of aluminum.

With no voltage applied, all the ribbons are at the same level.

When a voltage is applied between a ribbon and the electrode on the silicon substrate, an electric force pulls the ribbon downward.

The array of ribbons acts as a diffraction grating.



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Holography

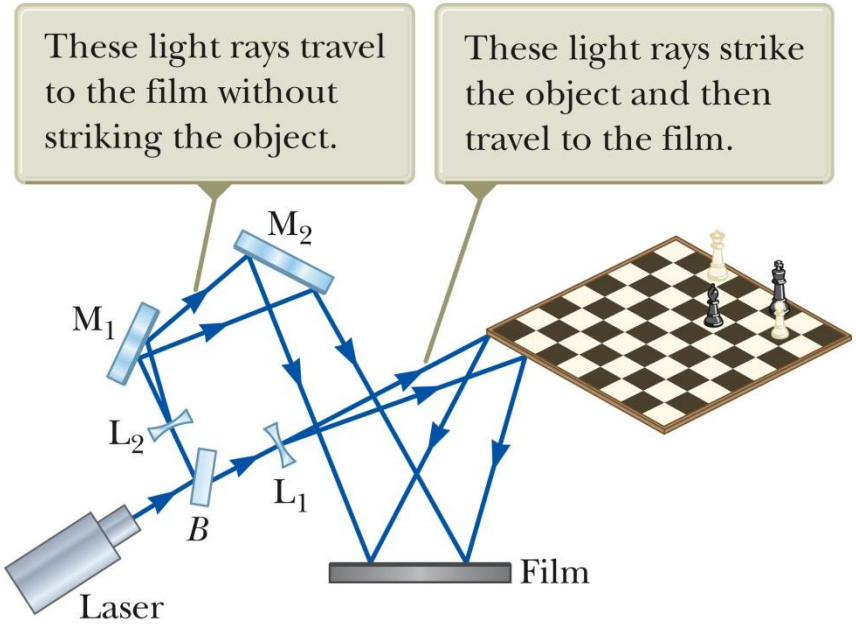
Holography is the production of three-dimensional images of objects.

Light from a laser is split into two parts by a half-silvered mirror at B.

One part of the light reflects off the object and strikes the film.

The other half of the beam is diverged by lens L_2.

It then reflects to mirrors M_1 and M_2 and then strikes the film.



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Holography, cont

The two beams overlap to form a complex interference pattern on the film.

The holograph records the intensity of the light reflected by the object as well as the phase difference between the reference bean and the beam scattered from the object.

Because of the phase difference, an interference pattern is formed that produces an image in which all three-dimensional information available from the perspective of any point on the hologram is preserved.

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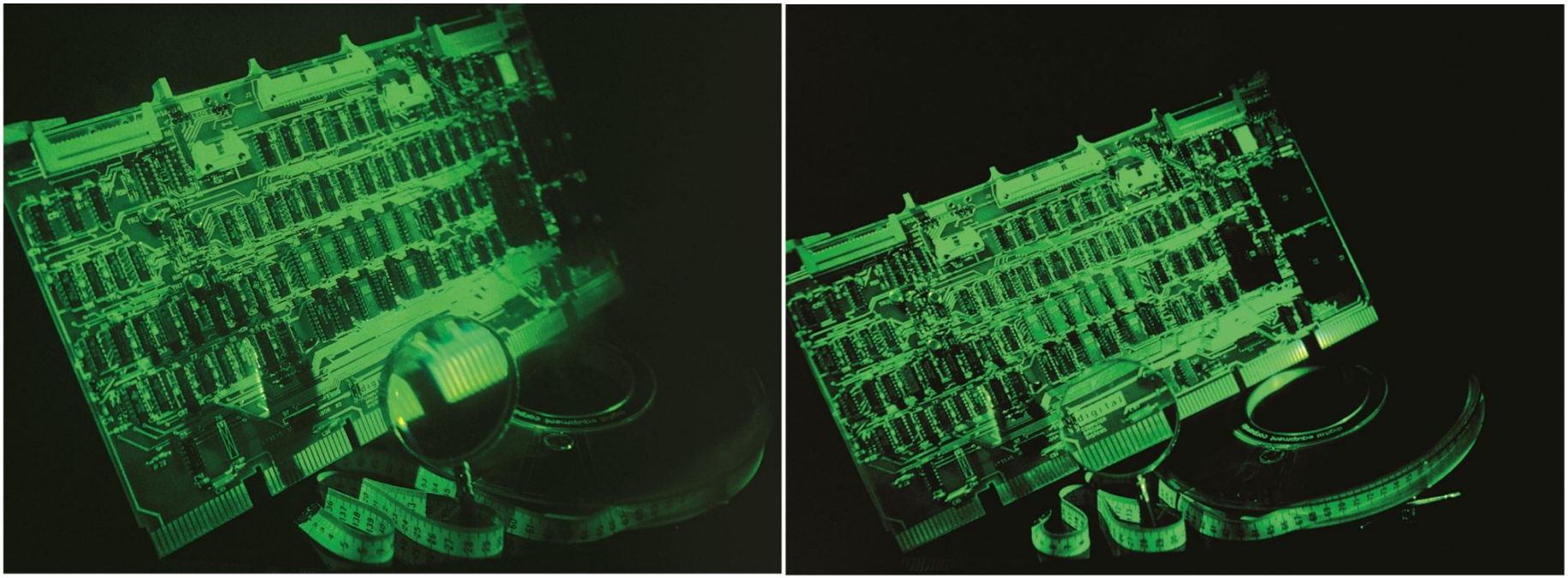
Holography, Example



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Diffraction of X-Rays by Crystals

X-rays are electromagnetic waves of very short wavelength.

Max von Laue suggested that the regular array of atoms in a crystal could act as a three-dimensional diffraction grating for x-rays.

Subsequent experiments confirmed this prediction.

The diffraction patterns from crystals are complex because of the three-dimensional nature of the crystal structure.

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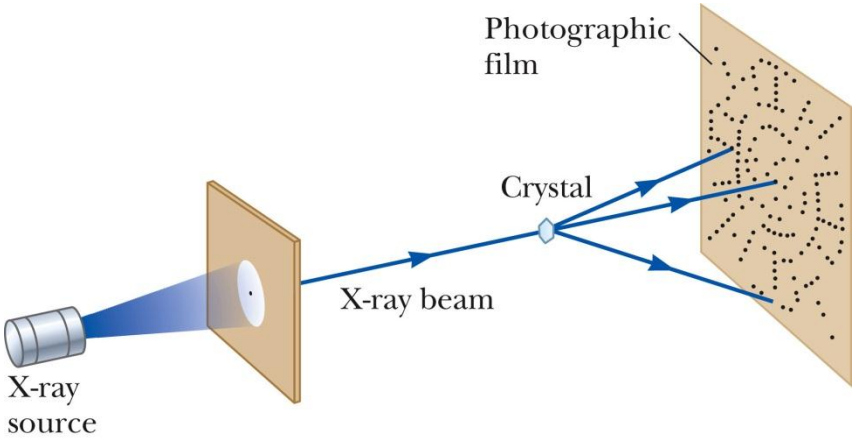
Diffraction of X-Rays by Crystals, Set-Up

A collimated beam of monochromatic x-rays is incident on a crystal.

The diffracted beams are very intense in certain directions.

- This corresponds to constructive interference from waves reflected from layers of atoms in the crystal.

The diffracted beams form an array of spots known as a *Laue pattern.*



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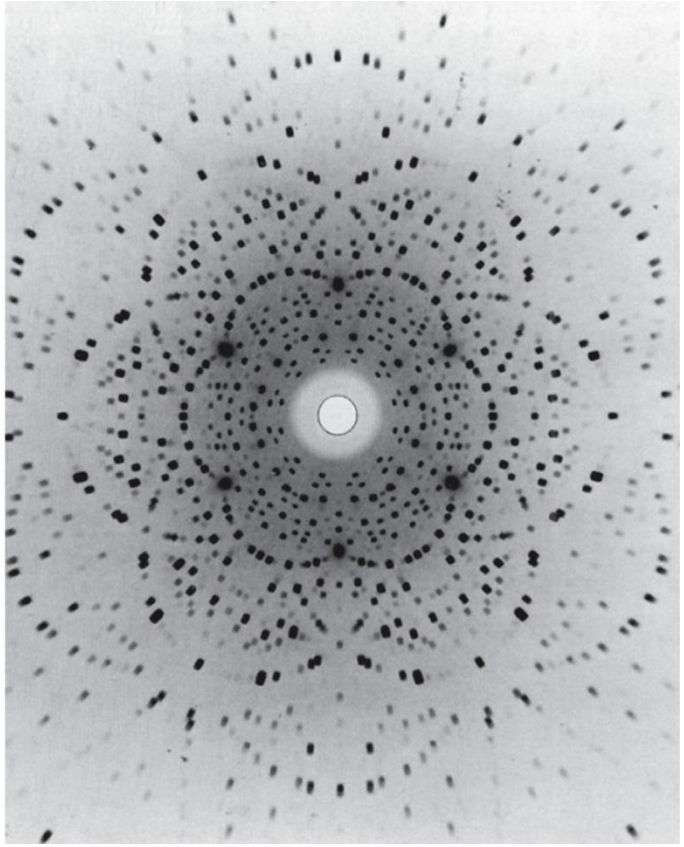
Laue Pattern for Beryl



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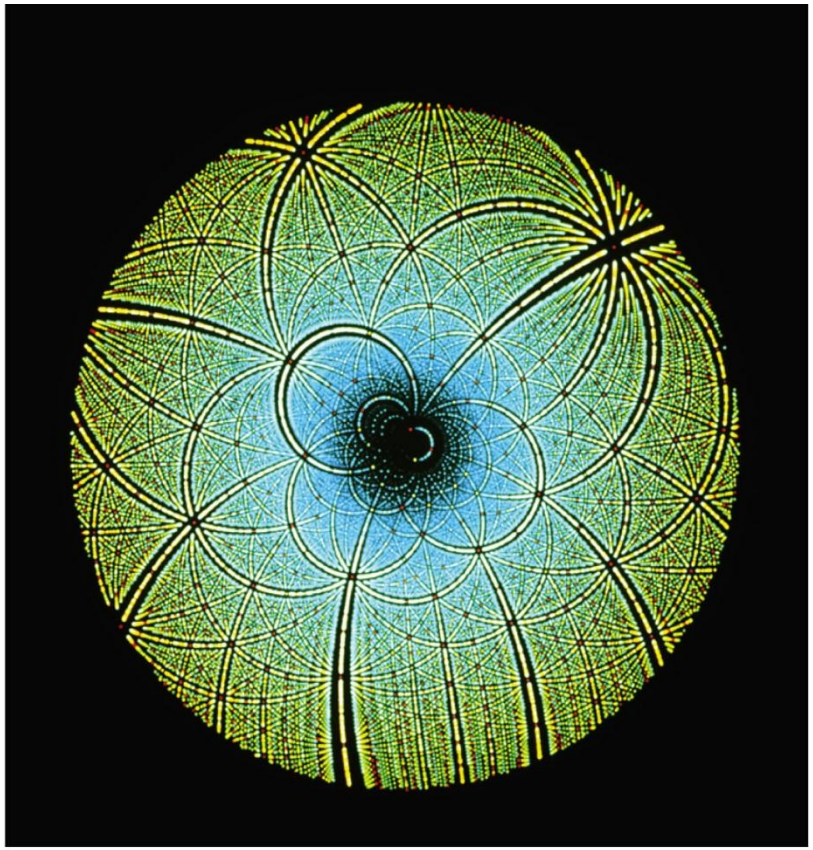
Laue Pattern for Rubisco



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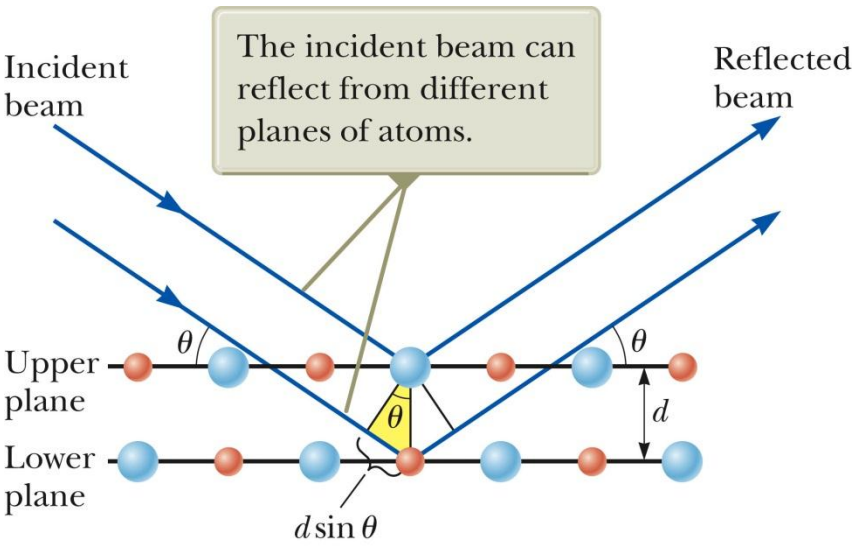
X-Ray Diffraction, Equations

This is a two-dimensional description of the reflection of the x-ray beams.

The condition for *constructive interference* is $2d \sin \theta = m\lambda$ where $m = 1, 2, 3$

This condition is known as **Bragg's law.**

This can also be used to calculate the spacing between atomic planes.



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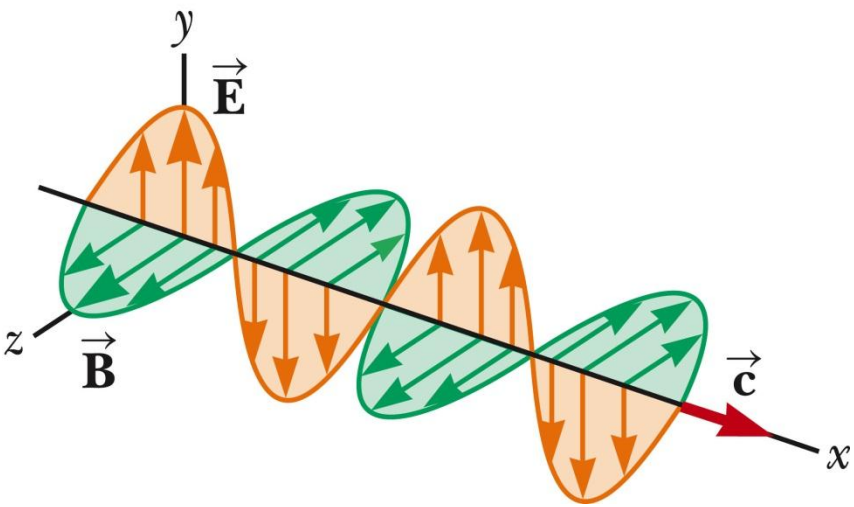
Polarization of Light Waves

The *direction of polarization* of each individual wave is defined to be the direction in which the electric field is vibrating.

In this example, the direction of polarization is along the *y*-axis.

All individual electromagnetic waves traveling in the *x* direction have an electric field vector parallel to the *yz* plane.

This vector could be at any possible angle with respect to the *y* axis.



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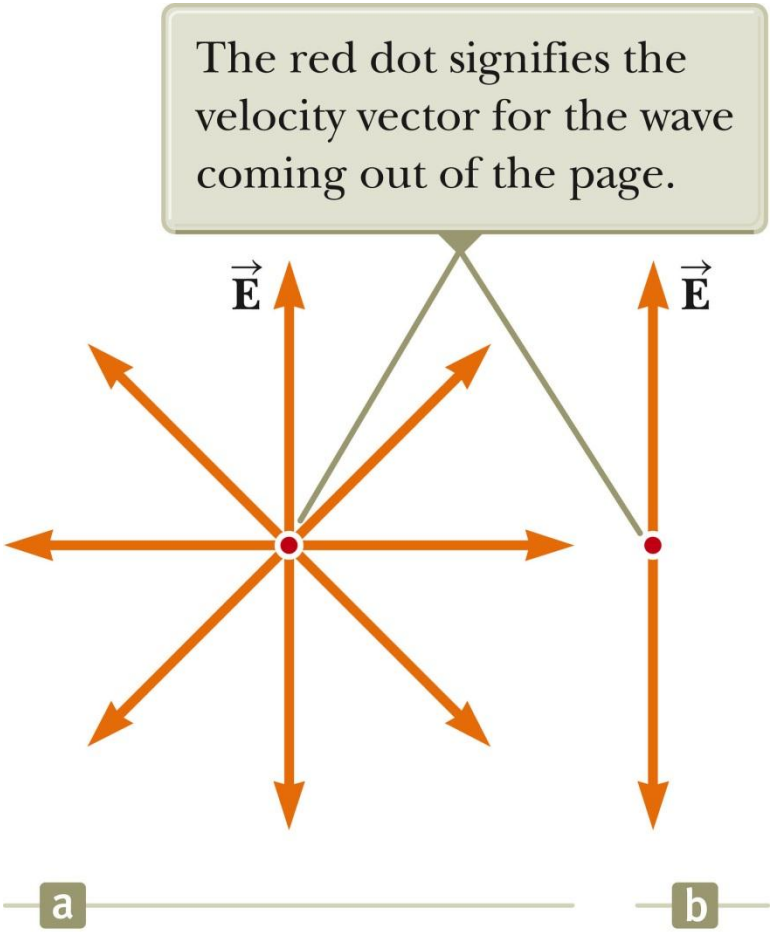
Unpolarized Light, Example

All directions of vibration from a wave source are possible.

The resultant em wave is a superposition of waves vibrating in many different directions.

This is an unpolarized wave.

The arrows show a few possible directions of the waves in the beam.



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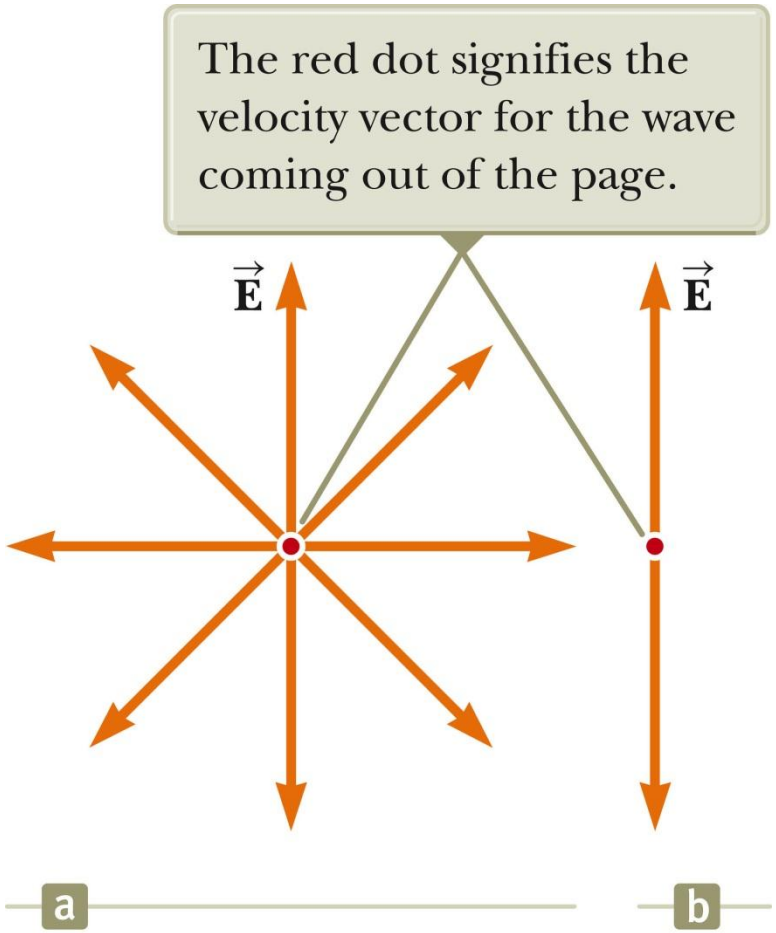
Polarization of Light, cont.

A wave is said to be *linearly polarized* if the resultant electric field vibrates in the same direction *at all times* at a particular point.

The plane formed by the field and the direction of propagation is called the *plane of polarization* of the wave.



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Methods of Polarization

It is possible to obtain a linearly polarized beam from an unpolarized beam by removing all waves from the beam except those whose electric field vectors oscillate in a single plane.

Processes for accomplishing this include:

- Selective absorption
- Reflection
- Double refraction
- Scattering

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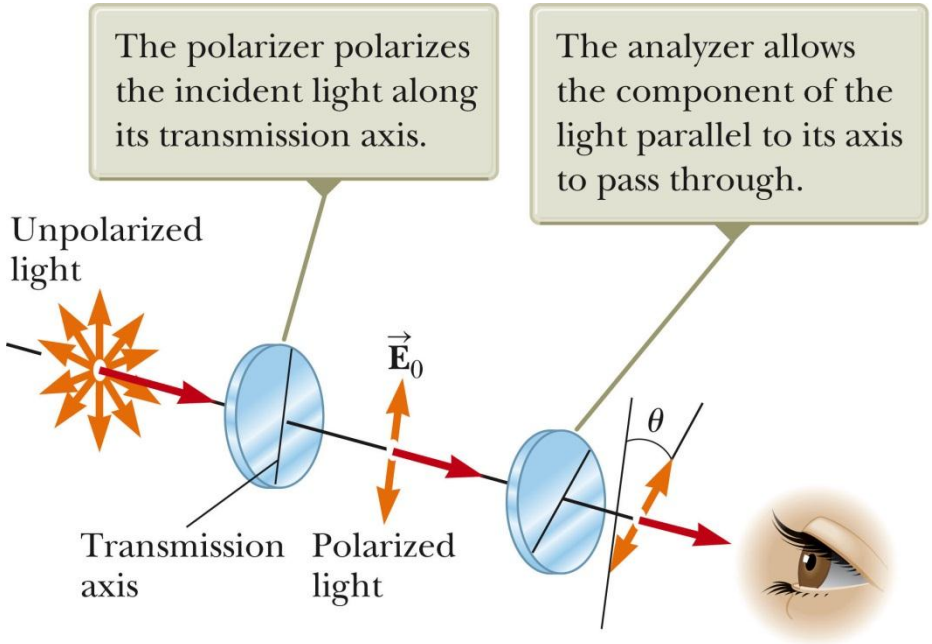
Polarization by Selective Absorption



The most common technique for polarizing light.

Uses a material that transmits waves whose electric field vectors lie in the plane parallel to a certain direction and absorbs waves whose electric field vectors are in all other directions.

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Selective Absorption, cont.

E. H. Land discovered a material that polarizes light through selective absorption.

- He called the material *Polaroid.*
- The molecules readily absorb light whose electric field vector is parallel to their lengths and allow light through whose electric field vector is perpendicular to their lengths.

It is common to refer to the direction perpendicular to the molecular chains as the *transmission axis.*

In an *ideal* polarizer,

- All light with the electric field parallel to the transmission axis is transmitted.
- All light with the electric field perpendicular to the transmission axis is absorbed.

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Intensity of a Polarized Beam

The intensity of the polarized beam transmitted through the second polarizing sheet (the analyzer) varies as

- $I = I_{max} \cos^2 \theta$
  - I_{max} is the intensity of the polarized wave incident on the analyzer.
  - This is known as **Malus' law** and applies to any two polarizing materials whose transmission axes are at an angle of θ to each other.

The intensity of the transmitted beam is a maximum when the transmission axes are parallel.

- $\theta = 0$ or 180°

The intensity is zero when the transmission axes are perpendicular to each other.

- This would cause complete absorption.

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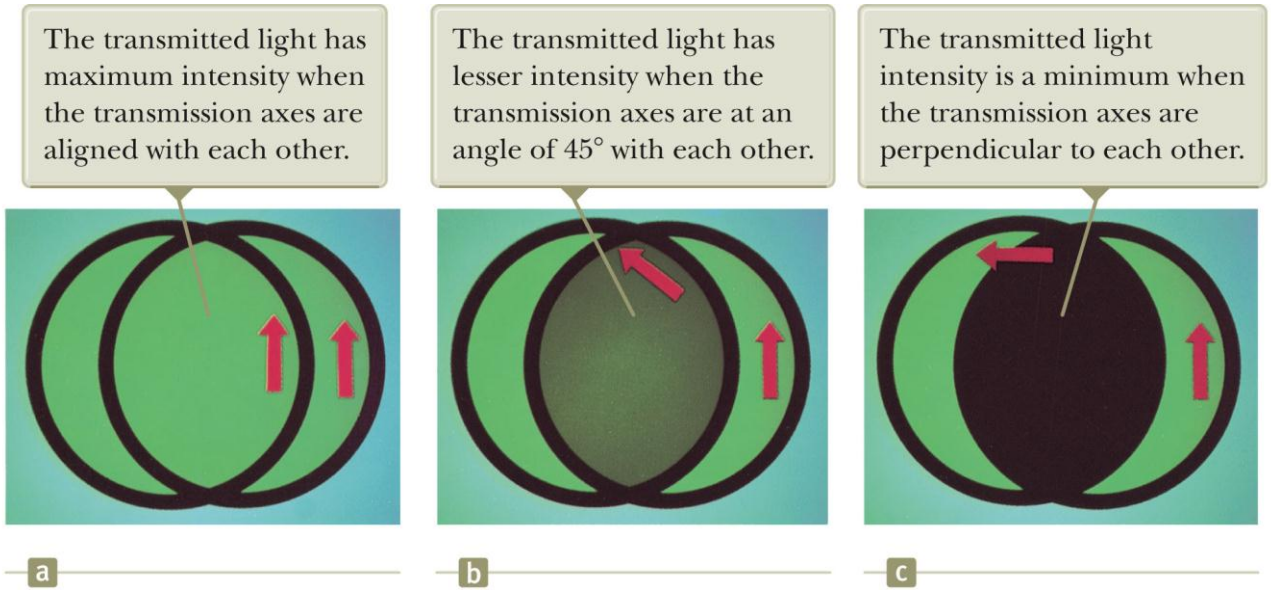
Intensity of Polarized Light, Examples



On the left, the transmission axes are aligned and maximum intensity occurs.

In the middle, the axes are at 45º to each other and less intensity occurs.

On the right, the transmission axes are perpendicular and the light intensity is a minimum.

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Polarization by Reflection

When an unpolarized light beam is reflected from a surface, the reflected light may be

- Completely polarized
- Partially polarized
- Unpolarized

The polarization depends on the angle of incidence.

- If the angle is 0°, the reflected beam is unpolarized.
- For other angles, there is some degree of polarization.
- For one particular angle, the beam is completely polarized.

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Polarization by Reflection, cont.

The angle of incidence for which the reflected beam is completely polarized is called the polarizing angle, θ_p.

Brewster's law relates the polarizing angle to the index of refraction for the material.

$$\tan \theta_p = \frac{n_2}{n_1}$$

θ_p may also be called Brewster's angle.

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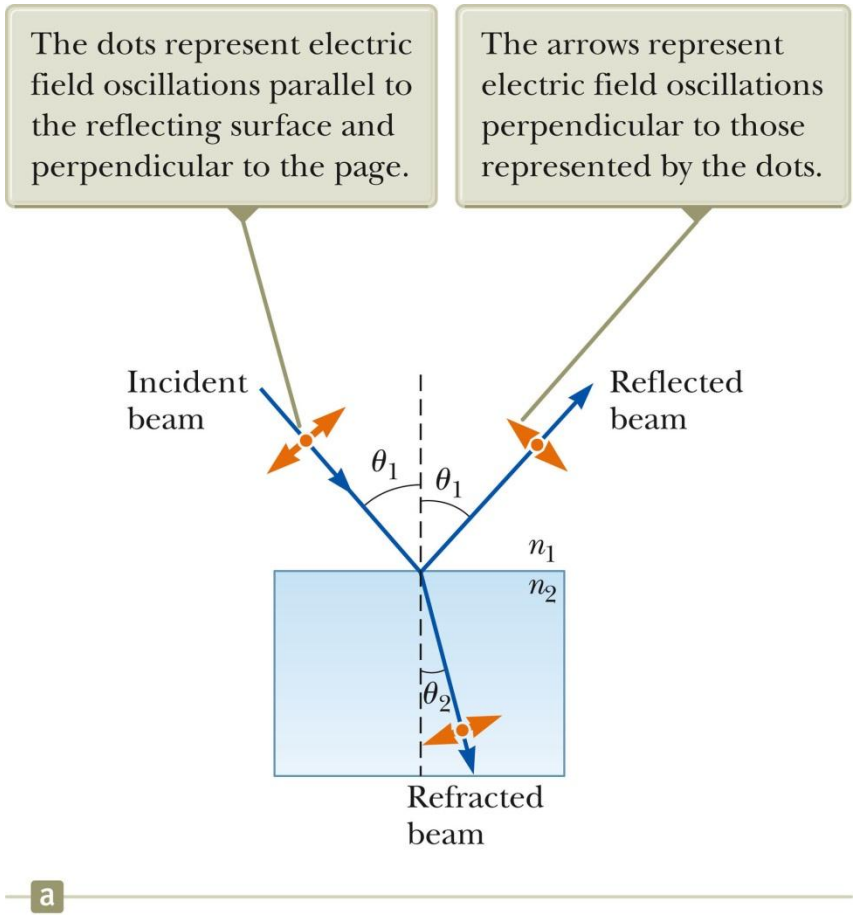
Polarization by Reflection, Partially Polarized Example

Unpolarized light is incident on a reflecting surface.

The reflected beam is partially polarized.

The refracted beam is partially polarized



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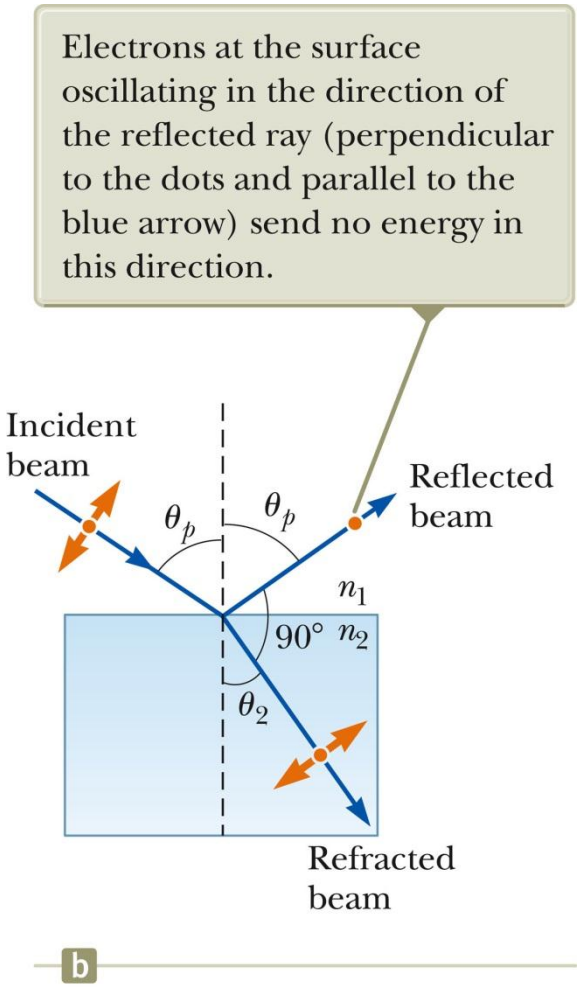
Polarization by Reflection, Completely Polarized Example

Unpolarized light is incident on a reflecting surface.

The reflected beam is completely polarized.

The refracted beam is perpendicular to the reflected beam.

The angle of incidence is Brewster's angle.



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Polarization by Double Refraction

In certain crystalline structures, the speed of light is not the same in all directions.

Such materials are characterized by two indices of refraction.

They are often called **double-refracting** or **birefringent** materials.

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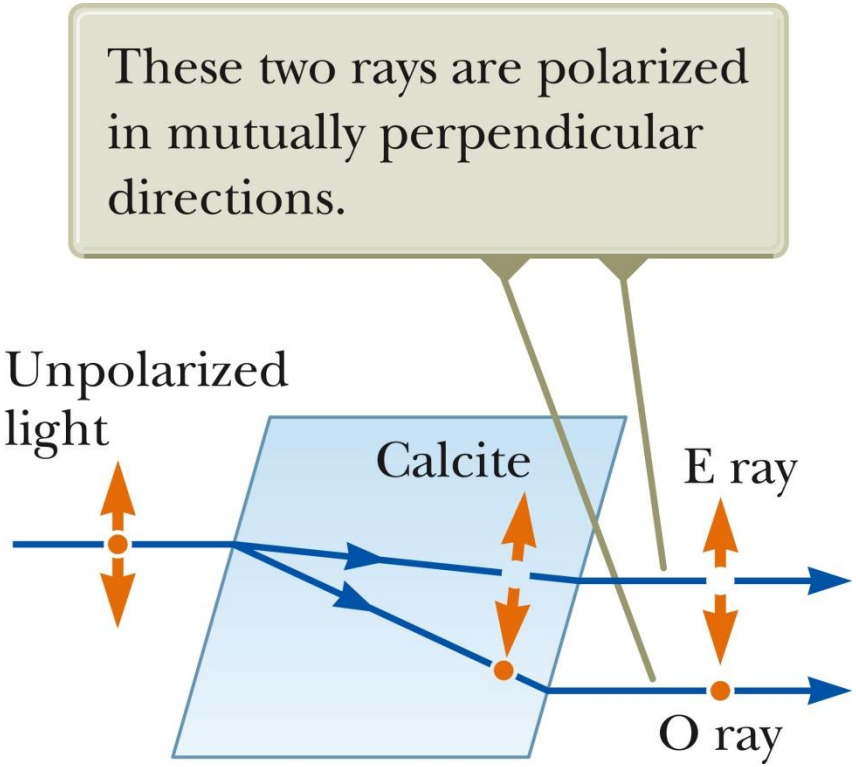
Polarization by Double Refraction, cont.

Unpolarized light splits into two plane-polarized rays.

The two rays are in mutual perpendicular directions.



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Polarization by Double Refraction, Rays

The **ordinary (O) ray** is characterized by an index of refraction of $n_{o.}$

- This is the same in all directions.

The second ray is the ***extraordinary (E) ray*** which travels at different speeds in different directions.

- Characterized by an index of refraction of n_E that varies with the direction of propagation.

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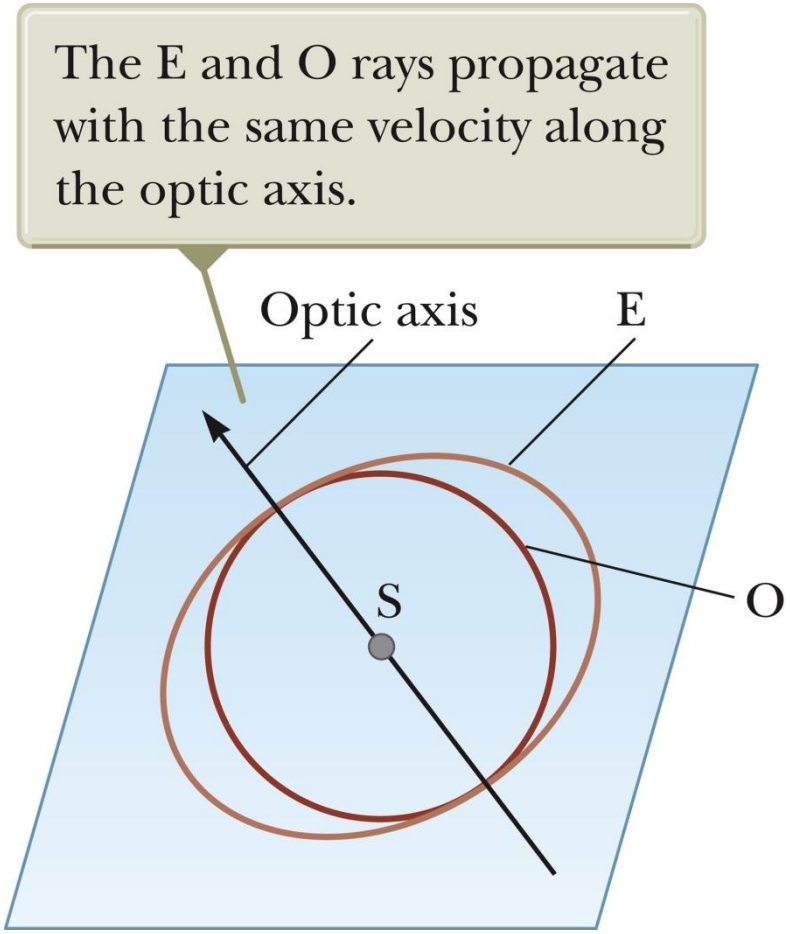
Polarization by Double Refraction, Optic Axis

There is one direction, called the **optic axis**, along which the ordinary and extraordinary rays have the same speed.

- $n_O = n_E$

The difference in speeds for the two rays is a maximum in the direction perpendicular to the optic axis.



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Some Indices of Refraction

TABLE 38.1 *Indices of Refraction for Some Double-Refracting Crystals at a Wavelength of 589.3 nm*

Crystal	n_O	n_E	n_O/n_E
Calcite ($CaCO_3$)	1.658	1.486	1.116
Quartz (SiO_2)	1.544	1.553	0.994
Sodium nitrate ($NaNO_3$)	1.587	1.336	1.188
Sodium sulfite ($NaSO_3$)	1.565	1.515	1.033
Zinc chloride ($ZnCl_2$)	1.687	1.713	0.985
Zinc sulfide (ZnS)	2.356	2.378	0.991

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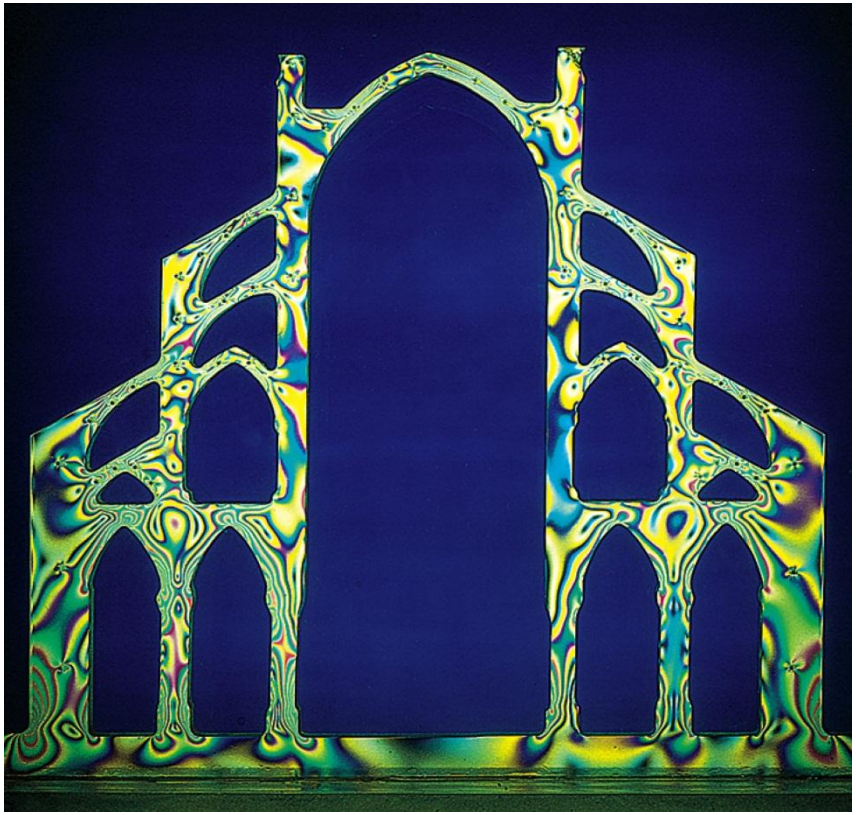
Optical Stress Analysis

Some materials become birefringent when stressed.

When a material is stressed, a series of light and dark bands is observed.

- The light bands correspond to areas of greatest stress.

Optical stress analysis uses plastic models to test for regions of potential weaknesses.



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Polarization by Scattering

When light is incident on any material, the electrons in the material can absorb and reradiate part of the light.

- This process is called **scattering.**

An example of scattering is the sunlight reaching an observer on the Earth being partially polarized.

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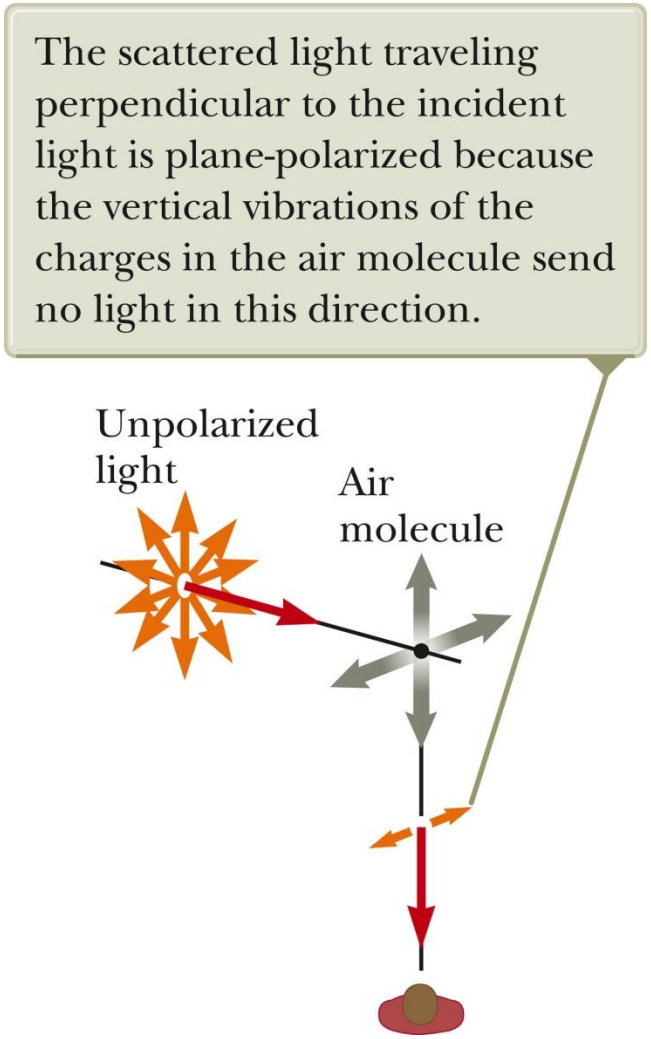
Polarization by Scattering, cont.

The horizontal part of the electric field vector in the incident wave causes the charges to vibrate horizontally.

The vertical part of the vector simultaneously causes them to vibrate vertically.

If the observer looks straight up, he sees light that is completely polarized in the horizontal direction.



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Scattering, cont.

Short wavelengths (violet) are scattered more efficiently than long wavelengths (red).

When sunlight is scattered by gas molecules in the air, the violet is scattered more intensely than the red.

When you look up, you see blue.

- Your eyes are more sensitive to blue, so you see blue instead of violet.

At sunrise or sunset, much of the blue is scattered away, leaving the light at the red end of the spectrum.

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Optical Activity

Certain materials display the property of **optical activity.**

- A material is said to be optically active if it rotates the plane of polarization of any light transmitted through it.
- Molecular asymmetry determines whether a material is optically active.

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